

39. 时变加速? 位变加速?



$$a_{\vec{r}} = \frac{\partial u_x}{\partial t} + \frac{\partial u_x}{\partial x} u_x + \frac{\partial u_x}{\partial y} u_y + \frac{\partial u_x}{\partial z} u_z$$

时变加速 位变加速

时变: 某一点上流速随时间的变化率。

位变: 某时刻 流速随位置的变化率。

$$a_{\vec{r}} = \frac{\partial u_x}{\partial t} + \frac{\partial u_x}{\partial x} u_x + \frac{\partial u_x}{\partial y} u_y + \frac{\partial u_x}{\partial z} u_z$$

40. 液体运动的基本形式?

① 平移

② 变形

③ 旋转

$$\omega_x = \frac{1}{2} \left(\frac{\partial u_y}{\partial z} - \frac{\partial u_z}{\partial y} \right) \dots$$

$$\omega_{xx} = \frac{\partial u_x}{\partial x} \dots$$

④ 剪切

$$\omega_y = \frac{1}{2} \left(\frac{\partial u_z}{\partial y} - \frac{\partial u_y}{\partial z} \right) \dots$$

